Supplementary material

Session-disjoint ECG authentication using rhythm-adaptive zone-cyclic morphology

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S1. Data organization and processing specification

The computational input comprises 36 arrays: 18 subjects with two recording blocks each. Array keys use the format subject_session, for example 002_S1. Each row is a mean-centered, zone-aligned cardiac cycle with 360 float64 coordinates. Values remain in the centered recording-amplitude scale, without unit-variance normalization. Subject identifiers are 001–003, 005–017 and 019–020. The original CEBSDB acquisition has a 5 kHz sampling rate; 360 denotes feature dimension rather than a sampling rate in hertz.

Each P–R, R–T and T–next-P slice is left-inclusive and right-exclusive. Its samples are linearly interpolated between their first and last indices to 48, 83 and 145 values, respectively. Concatenation gives 276 values. Linear interpolation at positions linspace(0, 275, 360) produces the final grid, and the row mean is subtracted. The resulting P, R and T reference phases are approximately 0, 48/275 and 131/275. The grid position masks for regional omission use $u_\ell = 275\,\ell / 359$: P–R is $u < 48$, R–T is $48 \le u < 131$, and T–next-P is $u \ge 131$.

The waveform-reading path performs channel mean subtraction, with division by 1,000 only when the centered maximum exceeds 1,000 input units. It does not implement an additional digital band-pass or notch filter. The project's landmark-generation code calls NeuroKit2 peak detection and delineation with library defaults corresponding to the NeuroKit and discrete-wavelet-transform methods, respectively. The historical library version is not independently established by the descriptor-level analysis. Missing timestamps are interpolated over beat index and rounded to four decimal places in seconds. Repeating the descriptor-level experiments does not require rerunning this detector; raw-landmark failure rates and full raw-to-descriptor benchmarking are outside the reported analysis.

All input arrays are finite. The maximum absolute row mean is 5.43×10^{-17}. Exact hashing found no repeated rows within a file and no equal rows across S1 and S2. Integrity checking does not establish physiological independence or absence of all acquisition-related dependence.

Table S1 lists unique cycle counts and the number of complete test descriptors after five-cycle grouping. A fitting/calibration split is performed before grouping and therefore has slightly different terminal remainders from grouping an entire test block.

Table S1. Unique aligned cycles and complete five-cycle test descriptors.

Subject	S1 cycles	S2 cycles	S1 K=5 probes	S2 K=5 probes
001	294	397	58	79
002	304	370	60	74
003	344	399	68	79
005	359	416	71	83
006	307	383	61	76
007	270	332	54	66
008	475	601	95	120
009	308	382	61	76
010	306	351	61	70
011	334	419	66	83
012	400	463	80	92
013	355	440	71	88
014	342	417	68	83
015	326	379	65	75
016	349	469	69	93
017	361	454	72	90
019	313	397	62	79
020	333	402	66	80
Total	6080	7471	1208	1486

S2. Reproducible fitting and scoring

For each claimant p, source block s and group length K, the deterministic seed is

$$r = 20261012 + 101\,p + 100011\{s = S2\} + 13\,K. \tag{S1}$$

For every identity separately, the fitting boundary is floor(0.7 * number_of_cycles). The preceding cycles form the fitting partition and the remaining cycles form the calibration partition. group retains the largest initial multiple of K and averages consecutive groups. No descriptor combines observations across partitions or subjects.

If the claimant provides n_g fitting descriptors and there are b negative-background identities, negative allocation is q, rem = divmod(n_g, b). Each sorted background identity contributes q randomly selected fitting descriptors, and the first rem identities contribute one additional descriptor. Sampling is without replacement. This produces exactly n_g negative fitting descriptors. A deterministic permutation then shuffles the balanced fitting set. The calibration and test sets retain all available negative descriptors from the specified identities.

Each supervised estimator is preceded by StandardScaler(with_mean=True, with_std=True), fitted only to the balanced fitting matrix. Label 1 is genuine and label 0 is impostor throughout training, scoring, confusion counts and genuine F1. Probabilities are explicitly selected for class 1. The

decision convention is score greater than or equal to threshold. No test-set fitting or probability recalibration is performed.

Table S2 gives the principal estimator settings. The file code/run_analysis.py defines the evaluated estimators under scikit-learn 1.8.0. Estimator defaults not listed below are those of that pinned version. All randomized estimators receive seed r. The reference Python and numerical-library versions are listed in requirements.txt; the script restricts numerical-library thread pools to one thread per job.

Table S2. Estimator specification. All supervised models use fitting-only StandardScaler.

Model	Explicit evaluation specification
SIC	Genuine mean and ddof=1 spread; K=1: κ=5.7, η=348/360; K=5: κ=8.4, η=328/360. Inclusive endpoints and coverage.
AdaBoost	50 depth-one trees; SAMME; learning_rate=1.
Neural network (MLP)	100 hidden units; ReLU; Adam; alpha=1; batch_size=auto; learning_rate_init=0.001; max_iter=1000; tol=0.0001; early_stopping=False.
Random forest	10 trees; max_depth=5; max_features=1; Gini; bootstrap=True; class_weight=None.
Decision tree	max_depth=5; Gini; splitter=best; class_weight=None.
Gaussian process	1.0 × RBF(1.0); fmin_l_bfgs_b; no optimizer restarts; max_iter_predict=100; copy_X_train=True.
Linear SVM	Linear kernel; C=0.025; probability=False; tol=0.001; class_weight=None.
Nearest neighbors	n_neighbors=3; weights=uniform; Minkowski p=2; algorithm=auto; leaf_size=30.

SIC uses genuine fitting observations only. The pointwise spread is calculated with ddof=1; membership includes both envelope endpoints. It uses $(\kappa,\eta)=(5.7,348/360)$ for $K=1$ and $(8.4,328/360)$ for $K=5$. Acceptance requires at least 348 or 328 accepted coordinates, respectively. Exact division is retained so that a threshold tie is accepted without a rounding-induced change. No stochastic-independence assumption converts the coverage score into a probability. The parameter pairs were chosen following exploratory assessment on the same cohort; their reported performance therefore requires independent validation and should not be interpreted as an unbiased estimate for prospectively specified parameters.

The executable package contains three stages: primary, secondary and unseen. Calculations are performed separately for each claimant, source block and K. The primary stage saves both calibration and test scores for all eight estimators. The secondary stage fits regional-omission controls, applies phase perturbations to fixed models, and performs the matched-count SIC experiment. The unseen stage exchanges the two negative-identity subsets. The SIC scores, fixed and calibrated decisions, and matched-count outputs are computed by code/run_analysis.py using the parameters listed above. Summarization is separate from fitting, so uncertainty estimates, tables and figures can be regenerated without refitting classifiers.

S3. Operating points, counts and empirical zero errors

The primary fixed thresholds are coverage $348/360$ for one-cycle SIC and $328/360$ for five-cycle SIC, decision score zero for linear SVM, and genuine-class probability 0.5 for all other models. SIC calibration retains the applicable κ_K and replaces only its coverage threshold. Thresholds selected from calibration use the complete empirical ROC candidate set. Minimizing calibration HTER resolves equal minima toward the highest threshold. The FAR-constrained alternative chooses the

highest genuine acceptance under the specified calibration FAR ceiling and uses the same conservative tie rule. It may choose an infinite threshold when rejecting all calibration samples is optimal or tied. Such an outcome is not silently replaced by a test-informed threshold.

Table S3 gives total primary confusion counts. A cycle can contribute impostor comparisons against multiple claimant verifiers; pooling the counts does not make those comparisons independent. For K = 1, the numbers of genuine and impostor comparisons are 13,551 and 230,367; for K = 5 they are 2,694 and 45,798. Counts are distinct from the unweighted mean rates in the main paper.

Table S3. Exact default-threshold confusion counts summed over both transfer directions and all claimants.

Classifier	K	TP	TN	FP	FN
SIC	1	11,832	225,187	5,180	1,719
SIC	5	2,457	44,025	1,773	237
AdaBoost	1	11,904	228,810	1,557	1,647
AdaBoost	5	2,313	45,264	534	381
Neural network (MLP)	1	13,188	229,801	566	363
Neural network (MLP)	5	2,648	45,691	107	46
Random Forest	1	11,637	228,224	2,143	1,914
Random Forest	5	2,143	45,583	215	551
Decision Tree	1	11,593	226,092	4,275	1,958
Decision Tree	5	2,220	44,368	1,430	474
Gaussian Process	1	7,531	226,880	3,487	6,020
Gaussian Process	5	2,616	45,745	53	78
Linear SVM	1	13,439	228,745	1,622	112
Linear SVM	5	2,682	45,531	267	12
Nearest neighbors	1	13,074	228,661	1,706	477
Nearest neighbors	5	2,604	45,493	305	90

For example, the fixed SVM yields 112 false rejections and 1,622 false acceptances for K = 1, and 12 and 267 for K = 5. SIC yields 1,719 false rejections and 5,180 false acceptances for K = 1, and 237 and 1,773 for K = 5. The pooled rates differ from unweighted claimant macro-means and are not substituted for them.

When a verifier has no observed errors of a given type in n comparisons, the independent-Bernoulli one-sided 95% upper reference bound is $1 - 0.05^{1/n}$. Repeated ECG comparisons are correlated, so this expression is an independence reference rather than a population guarantee for the present experiment. The per-verifier counts and full scores permit dependence-aware reanalysis. Likewise, a bootstrap interval with a zero endpoint does not establish that the population error rate is zero.

S4. Conditional uncertainty and paired comparisons

Each metric is averaged over directions within a claimant. For identity-holdout testing, folds are also averaged within claimant. A percentile bootstrap samples 18 claimant summaries with replacement, 20,000 times, using NumPy generator seed 271828. The 0.025 and 0.975 quantiles delimit the reported interval. No classifier is refitted inside the bootstrap. Consequently, these intervals

characterize the fixed fitted experiment conditional on its background cohort, not repeated recruitment and retraining of independent populations.

Paired K = 5 minus K = 1 HTER differences are computed within claimant before bootstrapping (table S4). The comparisons are exploratory and not multiplicity-adjusted. For SIC, K = 5 versus K = 1 changes both averaging and the parameter pair, so this contrast does not isolate the effect of averaging. An interval excluding zero does not establish superiority across datasets or classifier families.

Table S4. Paired difference in default-threshold HTER, K=5 minus K=1. Negative values favor K=5.

Classifier	**ΔHTER (percentage points)**	**95% claimant-bootstrap interval**
SIC	-0.751	[-3.518, +1.436]
AdaBoost	+2.002	[-1.886, +5.753]
Neural network (MLP)	-0.440	[-1.210, +0.060]
Random Forest	+2.853	[+0.501, +5.744]
Decision Tree	+1.598	[-3.717, +6.118]
Gaussian Process	-21.117	[-30.107, -12.259]
Linear SVM	-0.226	[-0.474, -0.004]
Nearest neighbors	-0.139	[-0.346, +0.107]

The area under the receiver operating characteristic curve (AUC) is computed from each verifier's continuous genuine score. EER linearly interpolates between adjacent empirical ROC points where FAR − FRR changes sign. The interpolation is descriptive: a fractional operating point need not correspond to an attainable discrete threshold, especially for nearest-neighbor or coverage scores. Figure S1 averages per-verifier operating curves at a common FAR grid, with equal claimant and direction weights. It does not pool uncalibrated scores from different verifiers.

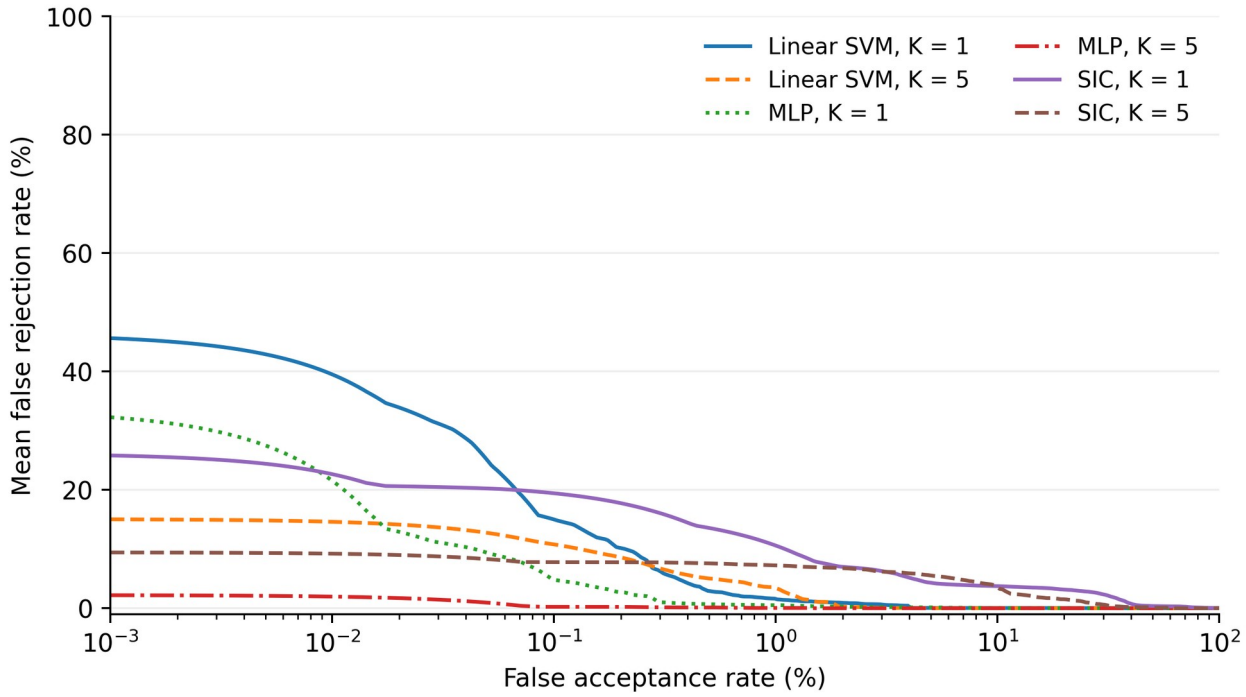


Figure S1. Descriptive test-score operating curves for selected models. Each verifier is interpolated onto a common FAR grid before equal-weight averaging over directions and claimants. The curves do not use pooled scores across independently fitted models and do not define operational thresholds. The lowest plotted FAR values may be below a single claimant's empirical resolution.

S5. Calibration transfer

Table S5 contrasts fixed and calibration-minimum-HTER operating points for selected classifiers. All calibration samples come from the final 30% of the enrollment block; the target block is never used to select the threshold. A low calibration error can coexist with a high test FRR if genuine scores shift between blocks. The conservative tie rule is part of the implemented method and must be retained when reproducing this comparison.

Table S5. Achieved test errors (%) at fixed and calibration-minimum-HTER thresholds. "Cal." denotes calibration-selected thresholds, not calibration-set error.

Classifier	K	Fixed HTER	Cal. HTER	Cal. FAR	Cal. FRR
SIC	1	6.421	6.916	0.151	13.680
SIC	5	5.670	15.508	0.002	31.014
Neural network (MLP)	1	1.335	6.705	0.038	13.371
Neural network (MLP)	5	0.895	11.744	0.004	23.483
Linear SVM	1	0.738	6.989	0.169	13.808
Linear SVM	5	0.513	13.577	0.121	27.033
Nearest neighbors	1	1.882	1.974	0.444	3.504
Nearest neighbors	5	1.743	1.666	0.337	2.994

Table S6 reports achieved test errors after targeting calibration FAR of 1% or 0.1%. The first column specifies the calibration target, not the observed test FAR. These results illustrate threshold transfer rather than guarantee conformance to a security requirement.

Table S6. Calibration FAR targets and achieved test errors (%). Targets are not guaranteed test constraints.

Classifier	K	Cal. FAR target	Test FAR	Test FRR	Test HTER
SIC	1	1	0.062	21.542	10.802
SIC	1	0.1	0.001	26.991	13.496
SIC	5	1	0.002	31.014	15.508
SIC	5	0.1	0.002	31.014	15.508
Neural network (MLP)	1	1	0.038	13.371	6.705
Neural network (MLP)	1	0.1	0.034	16.121	8.078
Neural network (MLP)	5	1	0.004	23.483	11.744
Neural network (MLP)	5	0.1	0.004	23.483	11.744
Linear SVM	1	1	0.153	13.832	6.992
Linear SVM	1	0.1	0.136	19.332	9.734
Linear SVM	5	1	0.119	27.033	13.576
Linear SVM	5	0.1	0.077	29.810	14.944
Nearest neighbors	1	1	0.444	3.504	1.974
Nearest neighbors	1	0.1	0.323	22.997	11.660
Nearest neighbors	5	1	0.337	2.994	1.666
Nearest neighbors	5	0.1	0.337	2.994	1.666

S6. Controlled sensitivity experiments

In the regional-omission experiment, the feature mask is applied to fitting, calibration and test arrays before fitting the scaler and classifier. The original full-cycle row centering is retained. Training examples and their ordering are identical to the full-feature experiment. The number of retained coordinates is 297 without P–R, 251 without R–T, and 172 without T–next-P. These dimensions follow directly from the grid mask rather than approximate allocation proportions.

For residual registration sensitivity, each test descriptor receives one deterministic pseudorandom displacement from a uniform distribution on [−a,a], with a equal to 1, 2, 4 or 8 final-grid samples. The displacement generator uses seed $r + 31a$. Sampling is linear, values beyond the endpoints use endpoint replication, and the resulting vector is re-centered. Perturbation is performed after five-cycle averaging so that both K settings experience residual error at the decision-input level. There is one perturbation realization per descriptor and magnitude, not a Monte Carlo estimate over many noise realizations. The result is not a raw landmark-detection accuracy test.

For the matched-count SIC experiment, source fitting cycles are divided into blocks of five. One envelope is fitted using the first cycle of each block and another using the block average. Both contain the same number of genuine fitting observations and use the same five-cycle test probes, $\kappa = 8.4$ and $\eta = 328/360$. Holding these parameters fixed separates the change in training representation from changes in sample count and decision tolerance. The experiment does not isolate variance from all other distributional differences between individual cycles and averages. Its results are presented in section S7.

In the identity-holdout experiment, sorted non-claimant identities are divided using alternating positions, not a search for an easy or difficult split. Two folds swap their roles, giving eight or nine training-background identities and nine or eight held-out identities. The genuine fitting and test observations are unchanged between the known/unseen comparison within a fold. The negative probes differ. No held-out impostor samples are used in fitting, scaling or calibration. The default threshold is used, and both folds are averaged in the reported diagnostic results.

S7. Matched-count SIC analysis

Table S7 and figure S2 compare envelopes fitted to individual cycles and five-cycle averages under the matched-count design in section S6. Both are evaluated on identical five-cycle test descriptors using $\kappa = 8.4$ and $\eta = 328 / 360$. The mean envelope width decreases from 0.225583 to 0.110320 centered recording units. The mean paired width ratio is 0.491 (95% claimant-bootstrap interval 0.465–0.520); ratios are computed within claimant and direction before averaging.

Table S7. Matched-count SIC results with common κ=8.4, η=328/360 and five-cycle test probes. Brackets show conditional 95% claimant-bootstrap intervals; widths use centered recording units.

Envelope fitted to	Mean width	FAR (%)	FRR (%)	HTER (%)
Individual cycles	0.225583 [0.193769, 0.258569]	39.468 [29.014, 49.546]	0.078 [0.000, 0.235]	19.773 [14.546, 24.794]
Five-cycle means	0.110320 [0.093578, 0.127335]	3.832 [0.953, 7.944]	7.508 [0.252, 20.096]	5.670 [1.409, 12.094]

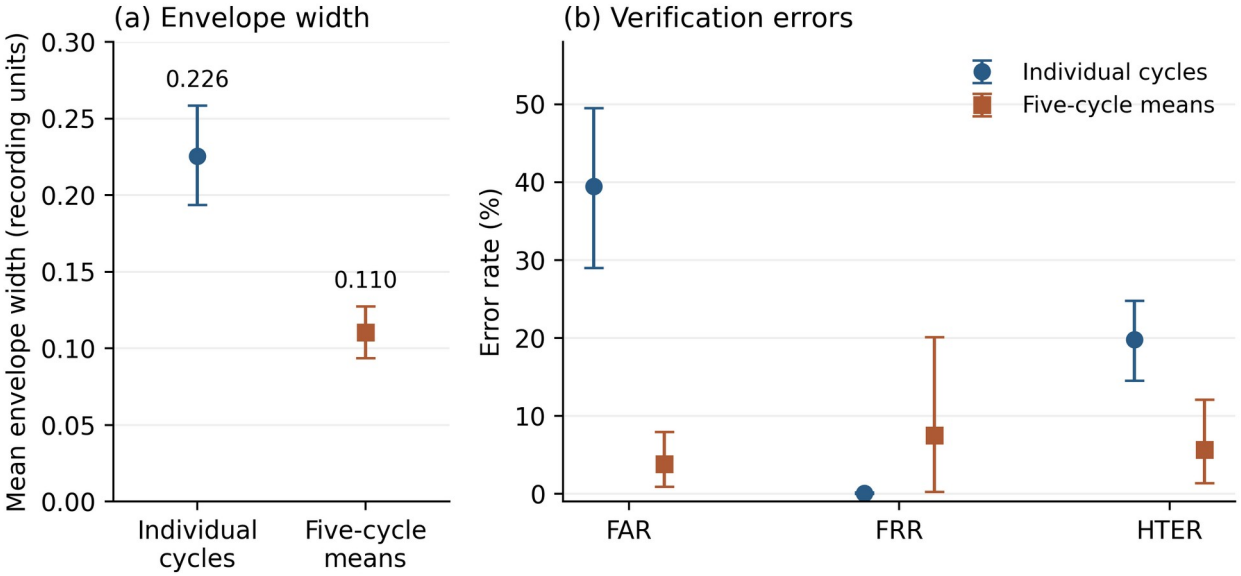


Figure S2. SIC fitting-representation control with equal genuine fitting counts, common κ=8.4 and η=328/360, and identical five-cycle test probes. (a) Mean envelope width. (b) FAR, FRR and HTER. Points are claimant-macro means and bars are conditional 95% claimant-bootstrap intervals. Individual-cycle fitting uses the first cycle of each five-cycle group.

The corresponding mean FAR falls from 39.468% to 3.832%, while mean FRR rises from 0.078% to 7.508%. Mean HTER decreases from 19.773% to 5.670%. The paired HTER difference is −14.103 percentage points (95% interval −20.784 to −6.420), and the paired FRR difference is +7.430 points (0.252–19.862). Thus, at this common operating point, the reduction in false acceptance outweighs the increase in genuine rejection.

This result characterizes how the training representation affects SIC under fixed fitting counts, test probes and tolerances. It is distinct from the primary comparison of K = 1 and K = 5, which also

changes the test representation and the parameter pair. For that primary comparison, the interval for the paired HTER difference spans zero (table S4). Neither contrast establishes an optimal averaging length or parameter pair.

S8. Data and code availability

The companion repository for the Python analysis code, aligned ECG cycle descriptors and numerical results is:

https://github.com/RomanButsiy/Supplementary_Material_Authentication_Data_and_Code

File paths below are relative to the analysis-package root. To reproduce the calculations, use the software versions specified in requirements.txt and the execution commands in the package README.md.

primary_all.csv contains claimant, direction, K, model, operating point, threshold, confusion counts, metrics and a fitting-duration diagnostic. Fitting durations are not controlled hardware benchmarks. results/scores.npz contains genuine-positive test labels (y), calibration labels (yv), each model's test and calibration scores, and SIC mean/spread vectors, width multipliers and coverage thresholds. secondary_all.csv and unseen_all.csv contain the controlled sensitivity and identity-holdout outputs. The summary_*.csv files retain full-precision means and conditional percentile limits; summary_equal_count_differences.csv contains the paired SIC width ratios and error differences. Score-array names use a claimant, source-block and K prefix, for example 002_S1_K5__, followed by the label, score or SIC-parameter name.

sample_counts.csv records cycle counts and fitting, calibration and test descriptor counts for each subject, block and K. The scripts code/run_analysis.py, code/summarize.py and code/make_figures.py reproduce model fitting, SIC scoring, numerical summaries and figures. The entry point code/run.py executes these steps and collects their outputs in a new directory. The analysis package uses data/cycles.npz as its input; all numerical result files and PDF figures are in results/.

The supplied descriptors contain no original timestamps or raw acquisition samples. They are sufficient to reproduce the reported descriptor-level experiments, but not to reinterpret phase displacements as milliseconds, evaluate raw-signal artifacts, or compare landmark detectors. Original source recordings remain available in CEBSDB. External ECG-ID validation, repeated sensor placement and mobile-device measurements are not included in these results.